✅ SmartTrack Developer Specification

✅ PART 1 — COMPLETE USER ANALYSIS (All Roles Identified)

SmartTrack is a **multi-tenant healthcare SaaS platform** with strict role isolation and PHI control. Below are ALL users identified across the documents and architecture:

🏢 1. ROOT LEVEL (Platform Owner Layer)

1. Super Admin (Root Owner)

Scope: Entire SmartTrack platform

Access: All organizations

Purpose:

- Platform configuration
- Global service codes
- Global payer rules
- Tenant management
- Audit oversight
- System-wide analytics

🏥 2. AGENCY LEVEL (Tenant Layer)

Each agency is isolated via RLS (Row Level Security).

2.1 Agency Admin

Scope: One agency only

Full administrative authority inside agency

Controls:

- Users & roles
- Programs
- Payers
- Rate tables
- Authorizations
- Compliance settings
- Notifications
- Billing thresholds

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2.2 Agency Supervisor

Scope: One agency

Clinical & Operational authority

Controls:

- Approve service records
- Review staff corrections
- Monitor schedules
- Review incidents
- Monitor authorization alerts

Cannot:

- Modify billing rates
- Access system-wide configs

2.3 Billing Specialist

Scope: Billing & Finance Console only

PHI visibility limited (no clinical narrative)

Controls:

- Pull approved service records
- Generate claims
- Submit EDI
- Track claim lifecycle
- Post 835 ERA payments
- Manage denials & appeals
- Enter service authorizations

Cannot:

- Modify clinical notes
- Approve service records

2.4 Finance Manager

Same as Billing Specialist but:

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- Write-off approval authority
- Financial reporting
- Payment reconciliation oversight

👩‍⚕️ 3. STAFF LEVEL

3.1 DSP (Direct Support Professional)

Primary frontline staff.

Controls:

- View assigned schedules
- Accept/Reject shifts
- Clock-in / Clock-out (EVV)
- Write daily notes
- Submit service record
- Request corrections

Cannot:

- Approve own records
- Access billing
- See other staff records

3.2 Clinical Staff (If applicable)

May:

- Enter assessments
- Upload documentation
- Contribute service documentation

Limited access similar to DSP but clinical module enabled.

👤 4. INDIVIDUAL / CLIENT (Optional Future Role)

Not direct system users currently, but:

- Portal may be introduced in future

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✅ PART 2 — COMPLETE END-TO-END WIREFRAMING FLOWS

Now let's define each user's **full journey from login to exit**, including every functional step.

🔐 SUPER ADMIN — END-TO-END WIREFRAME FLOW

1️⃣ Login

→ Email / Password
→ 2FA verification

2️⃣ Super Admin Dashboard

Layout Sections:

- Platform Metrics
  - Total Agencies
  - Total Users
  - Total Claims
  - Platform Revenue
- System Health Status
- Recent Agency Activity
- Global Alerts

3️⃣ Tenant Management

Flow:

- View Agency List
- Create New Agency
  - Agency Name
  - EIN
  - Billing Provider NPI
  - Address
- Activate / Suspend
- View Tenant Usage
- Audit Logs

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4️⃣ Global Service Code Management

- Add CPT/HCPCS code
- Add modifiers
- Define billing unit
- Assign default place of service
- Version tracking

5️⃣ Global Payer Rules

- Diagnosis required (Y/N)
- Rendering NPI required
- Claim type override
- Appeal window days

6️⃣ Audit & Compliance Monitoring

- View all org claim metrics
- Flag anomalies
- Fraud detection flags

🏥 AGENCY ADMIN — END-TO-END FLOW

1️⃣ Login

Role-based routing
→ /agency/dashboard

2️⃣ Dashboard Layout

Sections:

- Active Staff Count
- Upcoming Shifts
- Authorization Alerts
- Claims Status Snapshot

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- Compliance Score
- Incident Alerts

3️⃣ User & Role Management

Flow:

- Create User
- Assign Role
- Set Permissions
- Reset Password
- Deactivate User

4️⃣ Program Setup

- Create Program
  - Professional / Institutional
  - Billing type
- Link Service Codes

5️⃣ Payer Configuration

- Add payer
- Configure:
  - Payer ID
  - Clearinghouse
  - Diagnosis requirement
  - Rendering NPI requirement

6️⃣ Rate Tables

- Select payer
- Select service code
- Enter rate per unit
- Effective date range

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7️⃣ Authorization Monitoring

- View all authorizations
- Status (Active / Expiring / Exhausted)
- Drill into units used
- Receive alerts

👩‍⚕️ SUPERVISOR — END-TO-END FLOW

1️⃣ Dashboard

- Pending Service Records
- Authorization Alerts
- Staff Schedule Issues
- Missed EVV punches

2️⃣ Service Record Review

Flow:

- Open service record
- View:
  - Date
  - Units
  - Service Code
  - EVV GPS validation
  - Daily Note
- Approve / Reject
- Lock record

If Approved:
→ Status = APPROVED
→ Billing visible

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3️⃣ Schedule Oversight

- View accepted (Green)
- Rejected (Red)
- No Response (Gray)
- Reassign shift

4️⃣ Incident Review

- View incident report
- Approve / escalate

👩‍💼 BILLING SPECIALIST — END-TO-END FLOW

(Using document logic strictly)

1️⃣ Billing Dashboard

Sections:

- Unbilled Approved Records
- Claims in Draft
- Claims Submitted
- Denied Claims
- Payment Posted Today
- Authorization Alerts

2️⃣ Pull Approved Records

Query:

WHERE status = APPROVED

Filters:

- Date range
- Payer
- Program

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- Individual

Flags:

- No auth
- Auth near limit
- Rate not configured

3️⃣ Generate Claim

System performs:

1. Check coverage
2. Check rate table
3. Check authorization
4. Determine claim type (837P / 837I)
5. Create DRAFT claim
6. Build claim lines
7. Calculate total charge (cents)

4️⃣ Validation Layer

Errors:

- Missing NPI
- Missing EIN
- Zero units
- Duplicate claim
- Missing diagnosis

Cannot submit if:

- Blocking errors exist

5️⃣ Submit Claim

Option A: Clearinghouse API

Option B: Direct EDI

Creates:

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- claim_submission record

Status:
→ SUBMITTED

6️⃣ Track Claim Lifecycle

Statuses:

- DRAFT
- SUBMITTED
- ACCEPTED_277
- REJECTED_277
- PAID
- DENIED
- PARTIAL_PAYMENT
- ADJUSTED
- VOID
- APPEALED

7️⃣ Handle Denials

Actions:

- Appeal
- Correct & Resubmit
- Write Off
- Void

Mandatory reason logging.

8️⃣ Payment Reconciliation (835 ERA)

System:

- Parse ERA
- Match claim
- Post adjustments
- Update status
- Update units_used

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👩‍⚕️ DSP STAFF — END-TO-END FLOW

1️⃣ Login

2️⃣ My Schedule

Color Codes:

- Green = Accepted
- Red = Rejected
- Gray = No Action

3️⃣ Accept / Reject Shift

Immediate status update.

4️⃣ EVV Clock-In Flow

1. GPS validation
2. Time stamp
3. Confirm individual
4. Confirm location
5. Clock-in success screen

5️⃣ Service Documentation

After shift:

- Enter units delivered
- Write daily note
- Select service code
- Attach documentation

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Status:
→ DRAFT
→ Submit
→ PENDING_REVIEW

6️⃣ Correction Request

If missed punch:

- Submit correction request
- Supervisor review
- Audit logged

✅ PART 3 — DETAILED LAYOUT ARCHITECTURE (Per Role)

Each portal must have:

- Sidebar navigation
- Top header (notifications + profile)
- Role badge
- Tenant isolation

✅ PART 4 — HOW ALL MODULES CONNECT

This is the core architecture chain:
DSP → Service Record (DRAFT)
→ PENDING_REVIEW
→ Supervisor Approves
→ Status = APPROVED
→ Billing pulls record
→ Claim generated
→ Authorization units updated
→ Claim submitted
→ 835 received
→ Payment posted
→ Financial reports updated
Everything is event-driven.

🔐 Security Architecture

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- JWT contains org_id
- RLS enforces tenant isolation
- Billing cannot read clinical narrative
- All overrides logged
- Every status change logged

📊 COMPLETE FEATURE MATRIX

Includes:
✔ Multi-tenant architecture
✔ EVV compliance
✔ Service documentation lifecycle
✔ Authorization tracking
✔ Claim generation
✔ EDI submission
✔ ERA reconciliation
✔ Denial workflow
✔ Appeal tracking
✔ Write-offs
✔ Financial adjustments
✔ Alerts & notifications
✔ Role-based access
✔ Audit logs
✔ Threshold alerts
✔ Nightly jobs
✔ Status lifecycle automation

🎯 What This Means for Implementation

To build this correctly:

1. Backend first (data integrity)
2. RLS enforcement
3. Workflow triggers
4. Validation engine
5. Notification engine
6. EDI integration
7. Financial reconciliation layer

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This is a comprehensive care documentation and billing system designed for Intellectual and Developmental Disabilities (IDD) service providers, with a focus on strict HIPAA compliance and multi-console isolation.

1. User Roles & Admin Hierarchy

SmartTrack utilizes a strict **Role-Based Access Control (RBAC)** model enforced at the UI, API, and Database layers.

Platform-Level (Internal)

- **Platform Owner:** Highest authority (CTO/CEO); manages operator roles and initiates agency terminations.
- **Platform Admin:** Senior staff managing day-to-day agency approvals, subscriptions, and global settings.
- **Onboarding Specialist:** Guides new agencies through setup and configuration.
- **Support Engineer:** Investigates technical issues with restricted, approved "break-glass" read-only access.
- **Billing Operator:** Manages SmartTrack subscription billing (agency invoices).
- **Compliance Auditor:** Reviews platform-level audit logs for compliance.

Agency-Level (Customer)

- **Admin/Provider:** Agency owners/admins who manage users, programs, and organization-wide security.
- **Supervisor/Manager:** Program managers who review and approve staff documentation.
- **DSP (Direct Support Professional):** Front-line caregivers responsible for daily documentation and incident reporting.
- **Clinician:** Behavioral health specialists who create treatment plans and analyze behavior data.
- **Billing & Finance:** Specialists who process claims and reconcile payments.
- **Individual/Guardian:** Members or their families with read-only access to progress reports and schedules.

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2. End-to-End User Wireframes

Each console is a **standalone frontend application** to ensure zero data bleed.

Super Admin Console (Internal)

- **Dashboard:** Platform health metrics (uptime, API error rates, active agencies).

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- **Agency Pipeline:** A Kanban-style view showing signups moving from "Pending Review" to "Active".
- **Organization Directory:** Filterable list of all tenants; access to subscription and usage data.
- **Global Libraries:** Master lists for procedure codes and payers.

Admin Console (Agency)

- **User Management:** Table view for creating accounts and assigning sub-permissions.
- **Organization Config:** Screens for setting up programs, sites, and service codes.
- **Compliance Hub:** Exportable audit logs and session policy settings.

Supervisor Console

- **Review Queue:** List of pending notes from DSPs requiring approval or rejection.
- **ISP Monitoring:** Goal achievement charts and progress summary generators.
- **Incident Oversight:** Interface to review, comment on, and lock incident reports.

DSP Console

- **Individual Selection:** Only shows individuals explicitly assigned to the staff member.
- **Documentation Forms:** Structured fields for daily notes, shift logs, and ISP trials.
- **MAR Module:** Checklist for medication administration (if enabled).

Billing Console

- **Billing Queue:** Only shows **approved** service records.
- **Claim Builder:** Mapping interface for 837P/837I EDI generation.
- **Reconciliation:** Dashboard for matching 835 ERA files to submitted claims.

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3. Detailed Layout & Navigation

Common Layout Elements

- **Header:** Shows the current console name, user identity, and agency name.
- **Sidebar:** Distinct navigation links based on role (e.g., DSP sees "Daily Notes," Admin sees "User Privileges").
- **Footer:** HIPAA compliance badge and version number.

DSP Documentation Screen Layout

1. **Selection Header:** Individual's name and date.
2. **ISP Goals:** Checklist of goals to address during the shift.

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3. **Narrative Section:** Validated text box for daily observations.
4. **Action Bar:** "Save Draft" and "Submit for Review" (triggers electronic signature).

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4. Key Details & Interconnectivity

To achieve a seamless flow, the system relies on an **authoritative state machine** for service records:

How it Connects (The "Golden Thread")

1. **Assignment:** Admins map DSPs to Individuals in the `staff_assignments` table.
2. **Documentation:** DSPs create records; the status moves from `DRAFT` to `PENDING_REVIEW`.
3. **Approval:** Supervisors approve the record, which **locks** it (immutable) and sets status to `APPROVED`.
4. **Billing:** The Billing Console pulls only `APPROVED` records to generate claims.
5. **Audit Log:** Every step across every console is logged in an append-only table with user IDs and IP addresses.

Implementation Requirements

- **Encryption:** Field-level encryption for SSNs and Medicaid IDs; AES-256 for data at rest.
- **MFA:** Mandatory for all roles except DSP/Guardian (where it is configurable).
- **Tenant Isolation:** Row-Level Security (RLS) in PostgreSQL ensures one agency never sees another's data.

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